ECE 332: Engineering Electromagnetics II

Lecture 10

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Exam 1 Logistics

- Unless you have an accommodation, the exam will be held **in person, on paper** in FAB 48 from 1:30 to 3:10 p.m. on Thursday, May 7.
- If you have an **accommodation**, I will contact you separately to discuss details. If you follow the "Testing Services" link on the class Canvas site, you should be able to arrange to take the exam with your accommodation at the Testing Center.
- The exam will cover the material from **Chapters 6 & 7** of the text. **Main topics**:
  - applications of Faraday's and Ampére's laws,
  - charge-current continuity,
  - EM and time-harmonic potentials,
  - plane-wave propagation in lossless and lossy media,
  - polarization,
  - current flow in conductors,
  - EM power density.

Exam 1 Logistics (Continued)

- **I will provide space for your answers on the exam papers**. But, if you're worried about running out, feel free to bring some extra sheets of paper and a stapler (or other way to fasten papers together).
- The problems must **mostly be done by hand**.
  - A **calculator that is approved for taking the FE exam** (list of approved calculators) is allowed for arithmetic and evaluation of functions, such as trigonometric functions, exponential functions, and logarithms.
  - **Calculus must be done by hand**.
  - The **textbook or any other electronic devices** will **not be allowed**.
  - **I will provide any needed tables**.
- You'll be allowed **a two-sided, 1-page summary of important facts and formulas** to use during the exam. Aside from being a useful reference, the process of creating a summary can be helpful.
  I will not collect these.

Exam 1 Logistics (Continued)

- Please **show all work**. I can't give credit for work you don't show.
- **I will be in the room to answer any questions you might have during the exam**. Please come up to the lectern if you have a question. I will share any answers that could be beneficial with the rest of the class.
- I will help keep track of time by **displaying a clock** on the screen and **announcing a few time points**.
- **Comments/scores will be written on your exam papers**. Overall scores will be **posted in Canvas**. I will make an announcement in Canvas when the scores are available and provide some statistics.
- **Please remember to take slow, deep breaths and remember that there isn't enough time for me to ask you complicated questions.**

- The **4 vector fields E**, $\mathbf{D} = \epsilon\mathbf{E}$, **H**, and $\mathbf{B} = \mu\mathbf{H}$ describe electromagnetic phenomena and **determine the forces** that charges and currents experience.
- The effects of different sources can be evaluated separately and **superposed** (i.e. added) to determine the overall effect.
- These fields satisfy and are determined by **Maxwell's equations**.
- When any charges that aren't part of a current are **fixed in place** (electrostatics) and all currents are **steady** (magnetostatics), **Maxwell's equations are decoupled** and the equations describing the electric fields can be solved separately from the equations describing the magnetic fields.
- **Boundary conditions** specify the relationships between the normal and tangential components of the electromagnetic fields on the two sides of an interface between media with different electromagnetic properties.

Maxwell's Equations

- **When charges move and currents vary**, the equations for the electric fields are coupled to the equations for the magnetic fields and Maxwell's equations become

$$\nabla \cdot \mathbf{D} = \rho_v \longleftrightarrow \oint_S \mathbf{D} \cdot d\mathbf{s} = Q \quad \text{(Gauss' Law)} \tag{1}$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \longleftrightarrow \oint_C \mathbf{E} \cdot d\mathbf{l} = -\oint_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s} \quad \text{(Faraday's Law)} \tag{2}$$

$$\nabla \cdot \mathbf{B} = 0 \longleftrightarrow \oint_S \mathbf{B} \cdot d\mathbf{s} = 0 \quad \text{(Gauss' Law for Magnetism)} \tag{3}$$

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t} \longleftrightarrow \oint_C \mathbf{H} \cdot d\mathbf{l} = \oint_S \left(\mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}\right) \cdot d\mathbf{s} \quad \text{(Ampère's Law)} \tag{4}$$

In general, these would have be **solved simultaneously** to find the electromagnetic fields.

Faraday's Law

- The **magnetic flux** through a conducting loop with surface area S (in the direction of $d\mathbf{s}$) due to a magnetic field $\mathbf{B}$ is

$$\Phi = \int_S \mathbf{B} \cdot d\mathbf{s}. \tag{5}$$

- A change in the magnetic flux Φ induces a current in the loop. This is called **electromagnetic induction**.

- **Faraday's law** gives the **electromotive force (emf)**

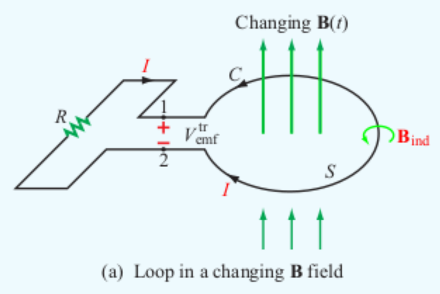
$$V_{\text{emf}} = -N\frac{d\Phi}{dt} = -N\frac{d}{dt}\int_S \mathbf{B} \cdot d\mathbf{s} \tag{6}$$

that the magnetic field $\mathbf{B}$ **induces** in a conducting loop with N turns.

- There are **3 cases** to consider.
  1. Time-varying flux through a stationary loop: V_{emf} is the **transformer emf** $V_{\text{emf}}^{\text{tr}}$.
  2. A loop moves in a stationary magnetic field: V_{emf} is the **motional emf** $V_{\text{emf}}^{\text{m}}$.
  3. A loop moves in a time-varying flux: $V_{\text{emf}} = V_{\text{emf}}^{\text{tr}} + V_{\text{emf}}^{\text{m}}$.

1. When the **loop is stationary** and the **magnetic field B changes**,

$$V_{\text{emf}} = V_{\text{emf}}^{\text{tr}}$$
$$= -N \int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s}. \quad (7)$$



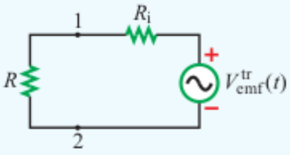
(a) Loop in a changing B field

- A **right-hand rule** specifies the polarity of V_{emf} and the direction of the contour C based on the direction of $d\mathbf{s}$.
- **Lenz's law** says that the induced current I and the polarity of V_{emf} will **oppose the change in magnetic flux** Φ. This determines the direction of the magnetic field $\mathbf{B}_{\text{ind}}$ associated with the induced current inside the loop.
  - $\mathbf{B}$ is increasing inside the loop $\implies$ $\mathbf{B}_{\text{ind}}$ points in the opposite direction from $\mathbf{B}$ inside the loop.
  - $\mathbf{B}$ is decreasing inside the loop $\implies$ $\mathbf{B}_{\text{ind}}$ points in the same direction as $\mathbf{B}$ inside the loop.
- The direction of $\mathbf{B}_{\text{ind}}$ determines the **direction of the induced current** I via the right-hand rule that specifies the relationship between a current and the associated magnetic field.

- If R_i is the resistance of the loop and R is the load on the circuit, the **equivalent circuit** satisfies

$$I = \frac{V_{\text{emf}}^{\text{tr}}}{R + R_i} \approx \frac{V_{\text{emf}}^{\text{tr}}}{R} \tag{8}$$

since, typically, $R_i \ll R$.



(b) Equivalent circuit

- We explored **a simple example** of an inductor in a changing magnetic field.

Review for Exam 1 (Continued)

- The relationship between the electric field $\mathbf{E}$ associated with the induced current and the external magnetic field $\mathbf{B}$ is the **integral form of Faraday's law**

$$\oint_C \mathbf{E} \cdot d\ell = -\int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s}. \tag{9}$$

- Stoke's theorem and the fact that the resulting equation must be true for any loop implies

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \tag{10}$$

which is the **differential form of Faraday's law**.

- For an **ideal transformer** that has a **primary coil** with N_1 turns and an ac voltage source $V_1(t)$ that creates a current I_1 and a **secondary coil** with N_2 turns, voltage V_2, current I_2, and a load R_L,

$$\frac{V_1}{V_2} = \frac{N_1}{N_2} \quad \text{and} \quad \frac{I_1}{I_2} = \frac{N_2}{N_1}. \qquad (11)$$

The **load on the primary coil** R_in is related to the **load on the secondary coil** R_L by

$$R_\mathrm{in} = \left(\frac{N_1}{N_2}\right)^2 R_\mathrm{L}, \quad Z_\mathrm{in} = \left(\frac{N_1}{N_2}\right)^2 Z_\mathrm{L}. \qquad (12)$$

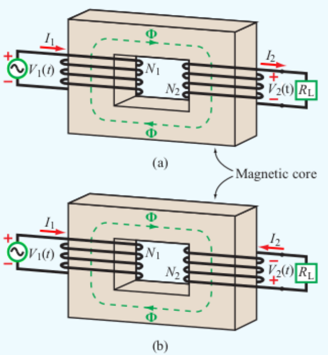


Figure 6-5 In a transformer, the directions of I_1 and I_2 are such that the flux Φ generated by one of them is opposite to that generated by the other. The direction of the secondary winding in (b) is opposite to that in (a), and so are the direction of I_2 and the polarity of V_2.

2. **When a closed circuit with contour C moves with velocity $\mathbf{u}$ through a static magnetic field $\mathbf{B}$**, then the **motional emf** is

$$V_{\text{emf}}^{\text{m}} = \oint_C (\mathbf{u} \times \mathbf{B}) \cdot d\boldsymbol{\ell}. \tag{13}$$

The integral is only evaluated along portions of the circuit that are moving and cross magnetic field lines. More precisely, if either **u** or **B** is parallel to $d\boldsymbol{\ell}$ for a part of the contour C, the integral will be 0 for that part of the contour.

- We explored an example of **a circuit containing a sliding bar**.

- We considered an example of finding the induced emf and current across a resistor in a **moving loop** in a **stationary non-uniform magnetic field**. Sides with components perpendicular to the plane of the velocity **u** and the magnetic field **B** contributed to the emf and those lying in the plane of the velocity and the magnetic field made no contribution.

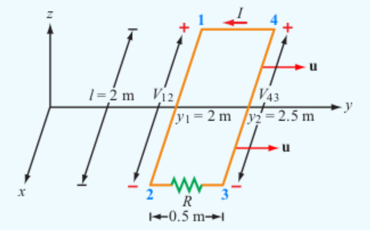


Figure 6-9 Moving loop of Example 6-4.

Review for Exam 1 (Continued)

- An **electromagnetic motor** or an **electromagnetic generator** consists of a conducting loop that rotates in a stationary magnetic field and are essentially the same device.
    - In a **motor**, the loop contains a voltage source that generate a torque and causes the loop to rotate. It converts electromagnetic power into mechanical power.
    - In a **generator**, a mechanical torque rotates the loop and a current is generated in the loop. It converts mechanical power into electromagnetic power.

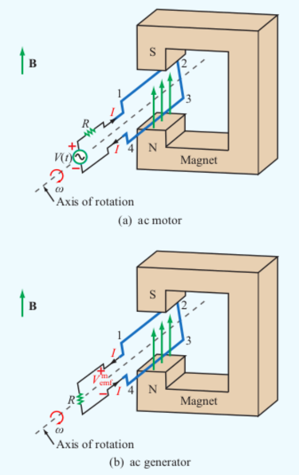


Figure 6-11 Principles of the ac motor and the ac generator. In (a) the magnetic torque on the wires causes the loop to rotate, and in (b) the rotating loop generates an emf.

Review for Exam 1 (Continued)

- **We considered the example of a generator** consisting of a rectangular loop with area A rotating at a constant angular velocity ω, starting from an angle C_0, in a constant magnetic field with magnitude B_0.
  - The **emf generated** in the loop is

$$V_{\mathrm{emf}}^{\mathrm{m}} = A\omega B_0 \sin(\omega t + C_0). \tag{14}$$

  - **We found this result in 2 ways –**
    1. Using the formula for $V_{\mathrm{emf}}^{\mathrm{m}}$ around the loop
    2. Using Faraday's law directly based on the flux Φ through the loop.
  - The formula can be used to **find ways to increase** the generated emf.

3. The 3rd **and most general case of applying Faraday's law** is when the conducting loop moves and the magnetic field changes. In this case, the **total emf** can be partitioned into **a transformer and a motional emf** and calculated as

$$V_{\text{emf}} = V_{\text{emf}}^{\text{tr}} + V_{\text{emf}}^{\text{m}} = -\int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s} + \oint_C (\mathbf{u} \times \mathbf{B}) \cdot d\boldsymbol{\ell}. \tag{15}$$

or **Faraday's law can be used directly** to determine

$$V_{\text{emf}} = -\frac{d\Phi}{dt} = -\frac{d}{dt}\int_S \mathbf{B} \cdot d\mathbf{s}, \text{ where } \Phi = \int_S \mathbf{B} \cdot d\mathbf{s}. \tag{16}$$

These are mathematically equivalent. Whichever approach is simpler will be used in a specific context.

- As an example of the most general case, we considered the **same generator with initial angle** $C_0 = 0$ **and a sinusoidally varying magnetic field**

$$\mathbf{B} = B_0 \cos(\omega t)\hat{\mathbf{z}}. \tag{17}$$

In this case, a direct application of Faraday's law via Φ showed that

$$V_{\text{emf}} = A\omega B_0 \sin(2\omega t), \tag{18}$$

which was very similar to the previous example where the magnetic field was constant.

Ampère's Law

- Faraday's law is one of Maxwell's equations that is different when charges move and magnetic fields change. The other equation that is different in the dynamic case is **Ampère's law**

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}. \tag{19}$$

- For an open surface S with boundary contour C, the **integral form of Ampère's law** can be written as

$$\oint_C \mathbf{H} \cdot d\mathbf{l} = I = I_c + I_d, \tag{20}$$

where I is the **total current** flowing through the surface,

$$I_c = \int_S \mathbf{J} \cdot d\mathbf{s} \tag{21}$$

is the **conduction current** flowing through the surface, and the **displacement current** is

$$I_d = \int_S \mathbf{J}_d \cdot d\mathbf{s}, \text{ where } \mathbf{J}_d = \frac{\partial \mathbf{D}}{\partial t}. \tag{22}$$

Here, $\mathbf{J}_d$ is the **displacement current density**.

- The example of an **ideal parallel-plate capacitor** showed that, even though the displacement current does not transport charges, it makes the total current in a circuit continuous.
- An example of a **more realistic wire** showed that I_d can generally be ignored in conductors.
- The **boundary conditions** pertaining to interfaces between materials with different electromagnetic properties have the same forms in the dynamic case as they did in the electrostatic and magnetostatic cases.

Review for Exam 1 (Continued)

Charge-Current Continuity Equation

- The **conservation of charge** implies the **charge-current continuity equation**

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho_v}{\partial t}. \tag{23}$$

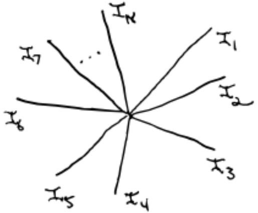
- In a conductor, $\nabla \cdot \mathbf{J} = 0$ and the charge-current continuity equation has the form

$$\oint_S \mathbf{J} \cdot d\mathbf{s} = 0, \tag{24}$$

which is **Kirchhoff's current law**.

- Kirchhoff's current law applied to a **junction in a circuit** where N branches (i.e. wires) carrying currents $I_1, I_2, \cdots, I_N$ meet has the form

$$\sum_{i=1}^{N} I_i = 0. \tag{25}$$



- **Excess charge in a conductor** emits an electric field that causes the electrons in the material to organize in ways that neutralize the interior of the conductor and give its surface the excess charge. This process follows the **charge-current continuity equation**, which takes the form

$$\frac{\partial \rho_v}{\partial t} + \frac{\sigma}{\epsilon}\rho_v = 0 \tag{26}$$

in the interior of the conductor. Thus, the **charge dissipates in the interior** according to the equation

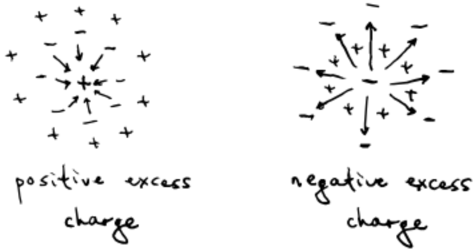
$$\rho_v(t) = \rho_{v0} e^{-(\sigma/\epsilon)t} = \rho_{v0} e^{-t/\tau_r} \quad (\mathrm{C/m^3}), \tag{27}$$

where $\rho_{v0} = \rho_v(0)$ is the charge density at $t = 0$ and

$$\tau_r = \frac{\epsilon}{\sigma} \tag{28}$$

is the **relaxation time constant**.

- The following diagrams illustrate how the electrons move.



- The **relaxation time constant** is **extremely small for conductors** (in copper, $\tau_r \sim 10^{-19}$ s) and **extremely large for good insulators** (in mica, $\tau_r \sim 10^{4}$ s).

Electric and Magnetic Potentials in the Dynamic Case

- When charges move and currents vary, the **scalar potential** V **cannot be defined by**

$$\mathbf{E} = -\nabla V \tag{29}$$

  since, in the dynamic version of Faraday's law, $\nabla \times \mathbf{E} \neq \mathbf{0}$.

- The **vector magnetic potential A**, which is defined by

$$\mathbf{B} = \nabla \times \mathbf{A}, \tag{30}$$

  is valid in the dynamic case since Gauss' law for magnetism, $\nabla \cdot \mathbf{B} = 0$, holds in both the dynamic and magnetostatic cases.

- In the dynamic case,

$$\mathbf{E} = -\nabla V - \frac{\partial \mathbf{A}}{\partial t} \tag{31}$$

  and

$$\mathbf{B} = \nabla \times \mathbf{A}. \tag{32}$$

Retarded Potentials

- **The scalar potential V at position $\mathbf{R}$** doesn't instantaneously respond to a change in the charge distribution. Instead, the effects of a change propagate to position $\mathbf{R}$ at a finite speed u_p. This means that the potential at $\mathbf{R}$ at time t depends on the position of the charge $\mathbf{R}_i$ at the previous time $t - R'/u_p$, where $R' = |\mathbf{R} - \mathbf{R}_i|$, i.e.

$$V(\mathbf{R}, t) = \frac{1}{4\pi\epsilon} \int_{\mathcal{V}'} \frac{\rho_v(\mathbf{R_i}, t - R'/u_p)}{R'} \, d\mathcal{V}'. \tag{33}$$

This is called the **retarded scalar potential**.

- Similarly, the vector potential **A** doesn't instantaneously change when currents change. The effects of such a change will propagate at the same speed u_p as the effects of a change in charge. Therefore, the vector potential at position **R** at time t is given by the **retarded vector potential**

$$\mathbf{A}(\mathbf{R}, t) = \frac{\mu}{4\pi} \int_{\mathcal{V}'} \frac{\mathbf{J}(\mathbf{R_i}, t - R'/u_p)}{R'} \, d\mathcal{V}'. \tag{34}$$

- The **propagation speed** u_p of the effects of changes in the charge distribution or changes in currents is essentially the **speed of light in the given medium**.

Time-Harmonic Potentials and Phasors

- The electric and magnetic fields $\mathbf{E}$ and $\mathbf{B}$ are **linearly dependent** on the scalar and vector potentials V and $\mathbf{A}$, which are **linearly dependent** on the charge and current densities ρ_v and $\mathbf{J}$. Therefore, the electric and magnetic fields are **linearly dependent** on the charge and current densities. So, it's possible to use **phasor domain methods** to determine the electric and magnetic fields corresponding to **sinusoidally-varying charge and current densities**.

- Charge and current densities $\rho_v(\mathbf{R_i}, t)$ and $\mathbf{J}(\mathbf{R_i}, t)$ with a complex sinusoidal time dependence are called **time-harmonic** and the corresponding potentials $V(\mathbf{R}, t)$ and $\mathbf{A}(\mathbf{R}, t)$ are called **time-harmonic potentials**.

- A **time-harmonic charge density** has the form

$$\rho_v(\mathbf{R_i}, t) = \rho_v(\mathbf{R_i}) \cos(\omega t + \phi) = \text{Re}\left\{\tilde{\rho}_v(\mathbf{R_i}) e^{j\omega t}\right\}, \tag{35}$$

where $\tilde{\rho}_v(\mathbf{R_i})$ is the **phasor** of $\rho_v(\mathbf{R_i}, t)$.

- The **phasor of the scalar potential** is related to the **phasor of the charge density** by

$$\tilde{V}(\mathbf{R}) = \frac{1}{4\pi\epsilon} \int_{\mathcal{V}'} \frac{\tilde{\rho}_v(\mathbf{R_i}) e^{-jkR'}}{R'} \, d\mathcal{V}'. \tag{36}$$

The **(real-valued) scalar potential** is

$$V(\mathbf{R}, t) = \text{Re}\left\{\tilde{V}(\mathbf{R}) e^{j\omega t}\right\} \tag{37}$$

and $k = \omega / u_p$ is the **wavenumber** or **phase constant**.

Review for Exam 1 (Continued)

- The **phasor of the vector potential** is related to the **phasor of the current density** by

$$\tilde{\mathbf{A}}(\mathbf{R}) = \frac{\mu}{4\pi} \int_{\mathcal{V}'} \frac{\tilde{\mathbf{J}}(\mathbf{R_i}) e^{-jkR'}}{R'} \, d\mathcal{V}'. \tag{38}$$

A **time-harmonic current density** has the form

$$\mathbf{J}(\mathbf{R}_i, t) = \mathbf{J}(\mathbf{R}_i) \cos(\omega t + \phi) = \operatorname{Re}\left\{\tilde{\mathbf{J}}(\mathbf{R}_i) e^{j\omega t}\right\} \tag{39}$$

and the **(real-valued) vector potential** is

$$\mathbf{A}(\mathbf{R}, t) = \operatorname{Re}\left\{\tilde{\mathbf{A}}(\mathbf{R}) e^{j\omega t}\right\}. \tag{40}$$

- The **phasor of the magnetic field intensity $\tilde{\mathbf{H}}$** is related to the **phasor of the vector potential $\tilde{\mathbf{A}}$** by the equation

$$\tilde{\mathbf{H}} = \frac{1}{\mu} \nabla \times \tilde{\mathbf{A}}. \tag{41}$$

Review for Exam 1 (Continued)

- **Faraday's law in the phasor domain** is

$$\nabla \times \tilde{\mathbf{E}} = -j\omega\mu\tilde{\mathbf{H}} \quad \text{or} \quad \tilde{\mathbf{H}} = -\frac{1}{j\omega\mu}\nabla \times \tilde{\mathbf{E}}. \tag{42}$$

- **Ampère's law in the phasor domain for a nonconducting medium** is

$$\nabla \times \tilde{\mathbf{H}} = j\omega\epsilon\tilde{\mathbf{E}} \quad \text{or} \quad \tilde{\mathbf{E}} = \frac{1}{j\omega\epsilon}\nabla \times \tilde{\mathbf{H}}. \tag{43}$$

- The above equations allow the determination of $\tilde{\mathbf{H}}$ given $\tilde{\mathbf{E}}$ or $\mathbf{H}$ given $\mathbf{E}$ in general, or the determination of $\tilde{\mathbf{E}}$ given $\tilde{\mathbf{H}}$ or $\mathbf{E}$ given $\mathbf{H}$ in a nonconducting medium.

- We explored **an example of using the phasor versions of Faraday's and Ampère's equations** to determine the magnetic field intensity $\mathbf{H}$ and the wavenumber k for a nonconducting medium when $\mathbf{E}$ is a specific time-harmonic field.

Electromagnetic (EM) Waves

- **Electromagnetic (EM) waves** occur due to the coupling of the electric and magnetic fields in Maxwell's equations.
- An **unbounded wave** travels through a homogeneous medium unimpeded, without encountering any boundaries.
- Media can be characterized as either **lossless** or **lossy**.
- In a **lossless medium**, $\sigma = 0$, EM waves are **never attenuated**, and no energy is dissipated.
- In a **lossy medium**, $\sigma \neq 0$, EM waves are **attenuated**, and energy is dissipated as heat.

Review for Exam 1 (Continued)

- Sources of EM waves generally emit waves with **spherical wavefronts**. A wavefront is a surface where the phase of the wave is constant.
- **“Far” from the source**, the wavefronts look **planar**.
- In a **uniform plane wave**, the fields are uniform on parallel planes.
- Our initial discussion was about **unbounded plane waves**.
- One way that EM waves can be bounded is when they're **guided**, which means that they **move along material structures**.
- **The confinement of radio waves in the HF band** between the ground and the ionosphere is an example of guided EM waves.
- **EM waves carried on a transmission line** is another example of guided EM waves. This also highlights the role of coupling between the electric and magnetic fields in the production of EM waves.

- The **phasor form of Maxwell's equations** is

$$\nabla \cdot \tilde{\mathbf{E}} = \tilde{\rho}_v/\epsilon \tag{44}$$

$$\nabla \times \tilde{\mathbf{E}} = -j\omega\mu\tilde{\mathbf{H}} \tag{45}$$

$$\nabla \cdot \tilde{\mathbf{H}} = 0 \tag{46}$$

$$\nabla \times \tilde{\mathbf{H}} = j\omega\epsilon_c\tilde{\mathbf{E}}, \tag{47}$$

where

$$\epsilon_c = \epsilon - j(\sigma/\omega) = \epsilon' - j\epsilon'' \tag{48}$$

is the **complex permittivity** and $\tilde{\mathbf{J}} = \sigma\tilde{\mathbf{E}}$. These describe the **spatial behavior** of the **time-harmonic fields**

$$\mathbf{E}(x, y, z, t) = \mathrm{Re}\left\{\tilde{\mathbf{E}}(x, y, z)e^{j\omega t}\right\}, \tag{49}$$

$$\mathbf{H}(x, y, z, t) = \mathrm{Re}\left\{\tilde{\mathbf{H}}(x, y, z)e^{j\omega t}\right\}, \tag{50}$$

$$\rho_v(x, y, z, t) = \mathrm{Re}\left\{\tilde{\rho_v}(x, y, z)e^{j\omega t}\right\}. \tag{51}$$

EM Waves in a Lossless, Source-Free, Unbounded Medium

- The phasor form of Maxwell's equations in a **source-free medium** (where $\tilde{\rho_v} = \rho_v = 0$) is

$$\nabla \cdot \tilde{\mathbf{E}} = 0 \tag{52}$$

$$\nabla \times \tilde{\mathbf{E}} = -j\omega\mu\tilde{\mathbf{H}} \tag{53}$$

$$\nabla \cdot \tilde{\mathbf{H}} = 0 \tag{54}$$

$$\nabla \times \tilde{\mathbf{H}} = j\omega\epsilon_c\tilde{\mathbf{E}}. \tag{55}$$

- In a **source-free medium**, the phasors of the electric and magnetic field intensities satisfy the homogeneous wave equations

$$\nabla^2\tilde{\mathbf{E}} - \gamma^2\tilde{\mathbf{E}} = 0 \tag{56}$$

$$\nabla^2\tilde{\mathbf{H}} - \gamma^2\tilde{\mathbf{H}} = 0, \tag{57}$$

where γ is the **propagation constant**, defined by

$$\gamma^2 = -\omega^2\mu\epsilon_c. \tag{58}$$

$\tilde{\mathbf{E}}$ and $\tilde{\mathbf{H}}$ obey the same equations and will have the same form.

- In a **lossless and source-free medium**, $\sigma = 0$, $\epsilon_c = \epsilon$, and the homogeneous wave equation for $\tilde{\mathbf{E}}$ is

$$\nabla^2 \tilde{\mathbf{E}} + k^2 \tilde{\mathbf{E}} = 0, \tag{59}$$

where the **wavenumber** is defined as

$$k = \omega\sqrt{\mu\epsilon}. \tag{60}$$

- In Cartesian coordinates,

$$\tilde{\mathbf{E}} = \tilde{E}_x \hat{\mathbf{x}} + \tilde{E}_y \hat{\mathbf{y}} + \tilde{E}_z \hat{\mathbf{z}} \tag{61}$$

and each component of $\tilde{\mathbf{E}}$ must satisfy the wave equation

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} + k^2 \right) \tilde{E}_* = 0, \tag{62}$$

where the asterisk can be replaced by x, y, or z.

Review for Exam 1 (Continued)

- A **uniform plane EM wave** has electric and magnetic fields that have **uniform properties on an infinite plane**. If the medium is **lossless** and **source-free** and (arbitrarily) **the plane is the x-y plane**, this means (These equations imply the wavefronts are $\parallel$ to the x-y plane.)

$$\frac{\partial \tilde{E}_x}{\partial x} = \frac{\partial \tilde{E}_x}{\partial y} = 0, \quad \frac{\partial \tilde{E}_y}{\partial x} = \frac{\partial \tilde{E}_y}{\partial y} = 0 \tag{63}$$

$$\frac{\partial \tilde{H}_x}{\partial x} = \frac{\partial \tilde{H}_x}{\partial y} = 0, \quad \frac{\partial \tilde{H}_y}{\partial x} = \frac{\partial \tilde{H}_y}{\partial y} = 0 \tag{64}$$

and the **wave equations** for $\tilde{E}_x$, $\tilde{E}_y$, $\tilde{H}_x$, $\tilde{H}_y$ are

$$\frac{d^2 \tilde{E}_x}{dz^2} + k^2 \tilde{E}_x = 0, \quad \frac{d^2 \tilde{E}_y}{dz^2} + k^2 \tilde{E}_y = 0, \tag{65}$$

$$\frac{d^2 \tilde{H}_x}{dz^2} + k^2 \tilde{H}_x = 0, \quad \frac{d^2 \tilde{H}_y}{dz^2} + k^2 \tilde{H}_y = 0, \tag{66}$$

where $\tilde{\mathbf{H}} = \tilde{H}_x \hat{\mathbf{x}} + \tilde{H}_y \hat{\mathbf{y}} + \tilde{H}_z \hat{\mathbf{z}}$.

- Under these circumstances, the z components of Ampère's law $\nabla \times \tilde{\mathbf{H}} = j\omega\epsilon\tilde{\mathbf{E}}$ and Faraday's law $\nabla \times \tilde{\mathbf{E}} = -j\omega\mu\tilde{\mathbf{H}}$ imply

$$\tilde{E}_z = \tilde{H}_z = 0, \tag{67}$$

which means **there are no electric or magnetic field components in the direction of propagation**.

- The **general solution** to

$$\frac{d^2\tilde{E}_x}{dz^2} + k^2\tilde{E}_x = 0 \tag{68}$$

is

$$\tilde{E}_x(z) = \tilde{E}_x^+(z) + \tilde{E}_x^-(z) = E_{x0}^+ e^{-jkz} + E_{x0}^- e^{jkz}, \tag{69}$$

where $\tilde{E}_x^+(z)$ is a wave with amplitude E_{x0}^+ in the $+z$ direction and $\tilde{E}_x^-(z)$ is a wave with amplitude E_{x0}^- in the $-z$ direction.

- The **phasor of the electric field intensity** for a **uniform plane wave** in the z direction with uniform properties on the x-y plane in a **lossless, source-free medium** has the form

$$\tilde{E}_x(z) = \tilde{E}_x^+(z) + \tilde{E}_x^-(z) = E_{x0}^+ e^{-jkz} + E_{x0}^- e^{jkz}, \tag{70}$$

where $\tilde{E}_x^+(z)$ is a wave with amplitude E_{x0}^+ in the $+z$ direction and $\tilde{E}_x^-(z)$ is a wave with amplitude E_{x0}^- in the $-z$ direction.

- Assuming $\tilde{E}_y = 0$ and $E_{x0}^- = 0$, the **phasor of the electric field** is

$$\tilde{\mathbf{E}}(z) = E_{x0}^+ e^{-jkz} \hat{\mathbf{x}} \tag{71}$$

and the **phasor of the magnetic field** is

$$\tilde{\mathbf{H}}(z) = H_{y0}^+ e^{-jkz} \hat{\mathbf{y}} = \frac{E_{x0}^+}{\eta} e^{-jkz} \hat{\mathbf{y}}, \tag{72}$$

where

$$\eta = \frac{\omega\mu}{k} = \sqrt{\frac{\mu}{\epsilon}}\ (\Omega). \tag{73}$$

Review for Exam 1 (Continued)

- The uniform plane wave is a **transverse electromagnetic wave (TEM)** since $\tilde{\mathbf{E}}$, $\tilde{\mathbf{H}}$, and the propagation direction are perpendicular.
- **Electric and magnetic field intensities** of the uniform plane wave:

$$\mathbf{E}(z,t) = \mathrm{Re}\left\{\tilde{\mathbf{E}}(z)e^{j\omega t}\right\} = |E_{x0}^{+}|\cos(\omega t - kz + \phi^{+})\hat{\mathbf{x}} \quad (\mathrm{V/m}) \tag{74}$$

$$\mathbf{H}(z,t) = \mathrm{Re}\left\{\tilde{\mathbf{H}}(z)e^{j\omega t}\right\} = \frac{|E_{x0}^{+}|}{\eta}\cos(\omega t - kz + \phi^{+})\hat{\mathbf{y}} \quad (\mathrm{A/m}), \tag{75}$$

where

$$E_{x0}^{+} = |E_{x0}^{+}|e^{j\phi^{+}}. \tag{76}$$

These fields are always **in phase** for **lossless media**.

- The **phase velocity** (speed of propagation) of the wave is

$$u_p = \frac{\omega}{k} = \frac{1}{\sqrt{\mu\epsilon}} \quad (\mathrm{m/s}), \tag{77}$$

i.e. the **speed of light** in the medium.

- The **wavelength** of the wave is

$$\lambda = \frac{2\pi}{k} = \frac{u_p}{f} \quad (\text{m}), \qquad (78)$$

where $f = \omega/2\pi$ is the **frequency**.

- The **intrinsic impedance of free space** is

$$\eta_0 = \sqrt{\frac{\mu_0}{\epsilon_0}} = 377 \quad (\Omega)$$
$$\approx 120\pi \ (\Omega). \qquad (79)$$

- We explored the example of an EM plane wave in air.

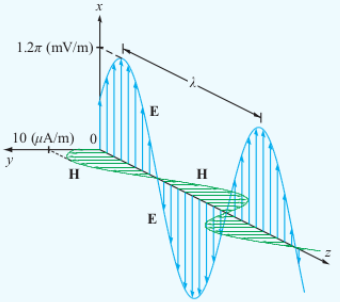


Figure 7-5 Spatial variations of **E** and **H** at $t = 0$ for the plane wave of Example 7-1.

- For a **uniform plane wave traveling in direction $\hat{\mathbf{k}}$ in any medium**

$$\tilde{\mathbf{H}} = \frac{1}{\eta}\hat{\mathbf{k}} \times \tilde{\mathbf{E}} \tag{80}$$

$$\tilde{\mathbf{E}} = -\eta\hat{\mathbf{k}} \times \tilde{\mathbf{H}}. \tag{81}$$

- The **directions of E, H, and $\hat{\mathbf{k}}$** are specified by a **right-hand rule** in which curling the fingers of the right hand from the direction of **E** to the direction of **H** causes the right thumb to point in the $\hat{\mathbf{k}}$ direction.

Examples:

- **Wave Propagating Along** $+z$ **(**$\hat{\mathbf{k}} = \hat{\mathbf{z}}$**) with E in** $\hat{\mathbf{x}}$ **Direction**

$$\tilde{\mathbf{E}} = E_{x0}^{+}e^{-jkz}\hat{\mathbf{x}} \tag{82}$$

$$\tilde{\mathbf{H}} = \frac{E_{x0}^{+}}{\eta}e^{-jkz}\hat{\mathbf{y}} \tag{83}$$

- **Wave Propagating Along** $-z$ **(**$\hat{\mathbf{k}} = -\hat{\mathbf{z}}$**) with E in** $\hat{\mathbf{x}}$ **Direction**

$$\tilde{\mathbf{E}} = E_{x0}^{-}e^{jkz}\hat{\mathbf{x}} \tag{84}$$

$$\tilde{\mathbf{H}} = -\frac{E_{x0}^{-}}{\eta}e^{jkz}\hat{\mathbf{y}} \tag{85}$$

- **Wave Propagating Along $+z$ ($\hat{\mathbf{k}} = \hat{\mathbf{z}}$) with E in $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$ Directions**

$$\tilde{\mathbf{E}} = E_{x0}^{+} e^{-jkz} \hat{\mathbf{x}} + E_{y0}^{+} e^{-jkz} \hat{\mathbf{y}} \tag{86}$$

$$\tilde{\mathbf{H}} = -\frac{E_{y0}^{+}}{\eta} e^{-jkz} \hat{\mathbf{x}} + \frac{E_{x0}^{+}}{\eta} e^{-jkz} \hat{\mathbf{y}} \tag{87}$$

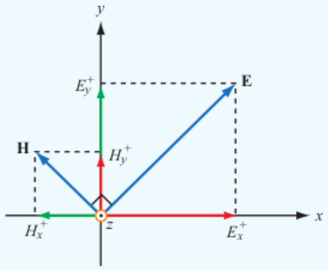


Figure 7-6 The wave $(\mathbf{E}, \mathbf{H})$ is equivalent to the sum of two waves, one with fields (E_x^+, H_y^+) and another with (E_y^+, H_x^+), with both traveling in the $+z$ direction.

Polarization

- The **polarization** of a uniform plane wave describes the curve that the electric field intensity vector $\mathbf{E}$ traces in a fixed plane perpendicular to the direction of propagation $\hat{\mathbf{k}}$.
- The curve is the one that the **phasor $\tilde{\mathbf{E}}$ follows**.
- The phasor has the form

$$\tilde{\mathbf{E}}(z) = E_{x0}e^{-jkz}\hat{\mathbf{x}} + E_{y0}e^{-jkz}\hat{\mathbf{y}}, \tag{88}$$

  where E_{x0} and E_{y0} are complex.
- The **magnitudes** $a_x = |E_{x0}|$ and $a_y = |E_{y0}|$ and the **phase difference** δ between the coefficients E_{x0} and E_{y0} **determine the polarization**. So, the phase of E_{x0} is defined to be 0 and the phase of E_{y0} is defined to be δ. This means

$$\tilde{\mathbf{E}}(z) = (a_x\hat{\mathbf{x}} + a_y e^{j\delta}\hat{\mathbf{y}})e^{-jkz} \tag{89}$$

and

$$\mathbf{E}(z,t) = a_x \cos(\omega t - kz)\hat{\mathbf{x}} + a_y \cos(\omega t - kz + \delta)\hat{\mathbf{y}}. \tag{90}$$

- The electric field intensity has **magnitude**

$$|\mathbf{E}(z,t)| = \left(a_x^2 \cos^2(\omega t - kz) + a_y^2 \cos^2(\omega t - kz + \delta)\right)^{1/2} \tag{91}$$

and has the **inclination angle**

$$\psi(z,t) = \tan^{-1}\left(\frac{E_y(z,t)}{E_x(z,t)}\right) \tag{92}$$

relative to the positive x axis, where $a_x = |E_{x0}|$ and $a_y = |E_{y0}|$.

- In general, $\mathbf{E}(z,t)$ follows an **ellipse**. But the ellipse can be a circle (**circular polarization**) or a line segment (**linear polarization**).

- **Linear Polarization**

  A uniform EM plan wave is linearly polarized when

  - $\delta = 0$ and $E_x(z,t)$ and $E_y(z,t)$ are **in phase**

    $$\mathbf{E}(z,t) = (a_x\hat{\mathbf{x}} + a_y\hat{\mathbf{y}})\cos(\omega t - kz) \quad (93)$$

    $$\psi(z,t) = \tan^{-1}\left(\frac{a_y}{a_x}\right) \quad (94)$$

  - $\delta = \pi$ and $E_x(z,t)$ and $E_y(z,t)$ are **out of phase**

    $$\mathbf{E}(z,t) = (a_x\hat{\mathbf{x}} - a_y\hat{\mathbf{y}})\cos(\omega t - kz) \quad (95)$$

    $$\psi(z,t) = \tan^{-1}\left(-\frac{a_y}{a_x}\right) \quad (96)$$

  In both cases,

  $$|\mathbf{E}(z,t)| = \sqrt{a_x^2 + a_y^2}\,|\cos(\omega t - kz)|. \quad (97)$$

- If $a_y = 0$, $\psi = 0$ or π and the wave is **x-polarized**.
- If $a_x = 0$, $\psi = \pi/2$ or $-\pi/2$ and the wave is **y-polarized**.

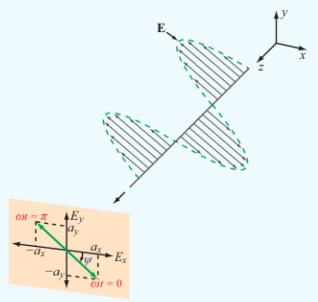


Figure 7-7 Linearly polarized wave traveling in the $+z$ direction (out of the page).

- **Circular Polarization**

  A uniform plane wave is circularly polarized when $a_x = a_y = a > 0$ and $\delta = \pm\pi/2$. In both cases, $|\mathbf{E}(z,t)| = a$.

  - **Left-Hand Circular (LHC) Polarization**

    In this case, $\delta = \pi/2$, which means

$$\tilde{\mathbf{E}}(z) = a(\hat{\mathbf{x}} + j\hat{\mathbf{y}})e^{-jkz}, \tag{98}$$
$$\mathbf{E}(z,t) = a(\cos(\omega t - kz)\hat{\mathbf{x}} - \sin(\omega t - kz)\hat{\mathbf{y}}), \tag{99}$$
$$\psi(z,t) = -(\omega t - kz). \tag{100}$$

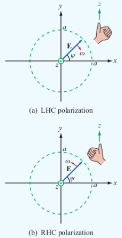


Figure 7-8 Circularly polarized plane waves propagating in the $+z$ direction (out of the page).

- **Right-Hand Circular (RHC) Polarization**

  In this case, $\delta = -\pi/2$, which means

$$\tilde{\mathbf{E}}(z) = a(\hat{\mathbf{x}} - j\hat{\mathbf{y}})e^{-jkz}, \tag{101}$$

$$\mathbf{E}(z,t) = a(\cos(\omega t - kz)\hat{\mathbf{x}} + \sin(\omega t - kz)\hat{\mathbf{y}}), \tag{102}$$

$$\psi(z,t) = \omega t - kz. \tag{103}$$

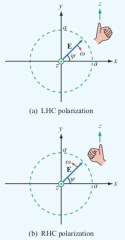


Figure 7-8 Circularly polarized plane waves propagating in the $+z$ direction (out of the page).

  For both types of circular polarization, **E** rotates in the opposite sense as a function of z for fixed t as it does as a function of t for fixed z.

- We explored an example of finding **E** and **H** for a RHC polarized wave.

- **Elliptical Polarization**

In this case,

$$\tilde{\mathbf{E}}(z) = (a_x\hat{\mathbf{x}} + a_y e^{j\delta}\hat{\mathbf{y}})e^{-jkz}, \quad (104)$$

where $a_x \neq a_y$ and $\delta \neq 0, \pm\pi/2, \pi$.

The ellipse that $\mathbf{E}(z,t)$ traces is specified by its

- **semi-major axis** a_ξ,
- **semi-minor axis** a_η,
- the **rotation angle** $\gamma \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ the major axis makes with a reference direction.

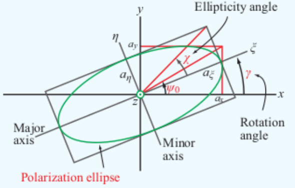


Figure 7-11 Polarization ellipse in the x–y plane, with the wave traveling in the z direction (out of the page).

- The **polarization is right- or left-handed**, following the same rules as for circular polarization.
- The **ellipticity angle** $\chi \in \left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$ specifies the **ratio between the major and minor axes** of the ellipse and the **handedness** of the polarization.
- The **axial ratio** $R = a_\xi / a_\eta$ satisfies

$$\tan\chi = \pm a_\eta / a_\xi = \pm 1/R. \quad (105)$$

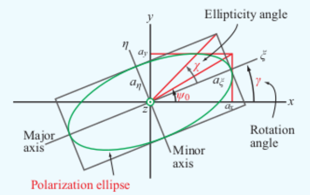


Figure 7-11 Polarization ellipse in the x–y plane, with the wave traveling in the z direction (out of the page).

- If $\tan\chi > 0$ (and $\chi > 0$), polarization is **left-handed**.
- If $\tan\chi < 0$ (and $\chi < 0$), polarization is **right-handed**.
- $R = 1$ (and $\chi = \pm\pi/4$) means the polarization is **circular**.
- $a_\eta = 0$ (i.e. $\chi = 0$, $R = \infty$) means the polarization is **linear**.

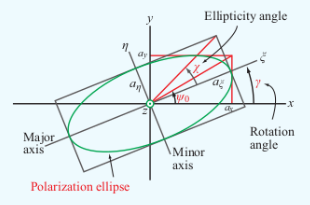


Figure 7-11 Polarization ellipse in the x–y plane, with the wave traveling in the z direction (out of the page).

- The **rotation angle** γ and **ellipticity angle** χ can be determined using a_x, a_y, and δ using

$$\tan 2\gamma = (\tan 2\psi_0) \cos\delta \tag{106}$$

$$\sin 2\chi = (\sin 2\psi_0) \sin\delta, \tag{107}$$

where

$$\psi_0 = \tan^{-1}\left(\frac{a_y}{a_x}\right) \in \left[0, \frac{\pi}{2}\right] \tag{108}$$

is the **auxiliary angle** and γ is required to have the same sign as $\cos\delta$. (Note: Knowing γ and χ doesn't completely specify the ellipse. Either the semi-major or semi-minor axis must also be specified.)

- Discussed an example of determining the polarization given **E**.

EM Plane Waves in a Lossy Medium

- The **phasor of a plane wave** propagating **in a lossy medium** satisfies the wave equation

$$\nabla^2 \tilde{\mathbf{E}} - \gamma^2 \tilde{\mathbf{E}} = \mathbf{0}, \tag{109}$$

where

$$\gamma^2 = -\omega^2 \mu \epsilon_c \quad \text{and} \quad \epsilon_c = \epsilon' - j\epsilon'' = \epsilon - j\frac{\sigma}{\omega}. \tag{110}$$

- The **attenuation constant** α is the real part of γ and the **phase constant** β is the imaginary part of γ. These are given by

$$\alpha = \omega \left\{ \frac{\mu\epsilon'}{2} \left[\sqrt{1 + \left(\frac{\epsilon''}{\epsilon'}\right)^2} - 1 \right] \right\}^{1/2} \ \text{(Np/m)} \tag{111}$$

$$\beta = \omega \left\{ \frac{\mu\epsilon'}{2} \left[\sqrt{1 + \left(\frac{\epsilon''}{\epsilon'}\right)^2} + 1 \right] \right\}^{1/2} \ \text{(rad/m)} \tag{112}$$

Review for Exam 1 (Continued)

- Assuming there is only a wave traveling in the $+z$ direction, the **electric field phasor satisfying the wave equation** is

$$\tilde{\mathbf{E}}(z) = E_{x0} e^{-\alpha z} e^{-j\beta z} \hat{\mathbf{x}} \tag{113}$$

and the corresponding **magnetic field phasor** is

$$\tilde{\mathbf{H}}(z) = \frac{E_{x0}}{\eta_c} e^{-\alpha z} e^{-j\beta z} \hat{\mathbf{y}}. \tag{114}$$

- The **intrinsic impedance for the lossy medium** is

$$\eta_c = \sqrt{\frac{\mu}{\epsilon_c}} = \sqrt{\frac{\mu}{\epsilon'}} \left(1 - j\frac{\epsilon''}{\epsilon'}\right)^{-1/2}, \tag{115}$$

where the **complex permittivity** is

$$\epsilon_c = \epsilon' - j\epsilon'' = \epsilon - j\frac{\sigma}{\omega}. \tag{116}$$

Since η_c is complex, $\tilde{\mathbf{E}}(z)$ and $\tilde{\mathbf{H}}(z)$ (and $\mathbf{E}(z,t)$ and $\mathbf{H}(z,t)$) **do not have the same phase**.

Review for Exam 1 (Continued)

- The **magnitudes**

$$|\tilde{\mathbf{E}}(z)| = |E_{x0}|e^{-\alpha z}, \quad |\tilde{\mathbf{H}}(z)| = \left|\frac{E_{x0}}{\eta_c}\right| e^{-\alpha z} \tag{117}$$

  decrease exponentially at a rate α and decrease by a factor of $1/e$ for each multiple of the **skin depth**

$$\delta_s = 1/\alpha \text{ m.} \tag{118}$$

  The lost energy is dissipated as heat.

- The **expressions for** α, β, and η_c are correct for any **linear, isotropic, homogeneous medium**.
- Media are characterized by the ratio ϵ''/ϵ':
  - $\epsilon''/\epsilon' \ll 1$ – **low-loss dielectric**
  - $\epsilon''/\epsilon' \gg 1$ – **good conductor**
  - $10^{-2} \leq \epsilon''/\epsilon' \leq 10^2$ – **quasi-conductor**

- **Low-Loss Dielectrics**

  Here, $\epsilon''/\epsilon' \ll 1$, which implies

$$\alpha \approx \frac{\omega\epsilon''}{2}\sqrt{\frac{\mu}{\epsilon'}} = \frac{\sigma}{2}\sqrt{\frac{\mu}{\epsilon}} \ \ (\mathrm{Np/m}), \tag{119}$$

$$\beta \approx \omega\sqrt{\mu\epsilon'} = \omega\sqrt{\mu\epsilon} \ \ (\mathrm{rad/m}), \tag{120}$$

$$\eta_c \approx \sqrt{\frac{\mu}{\epsilon}} \ \ (\Omega). \tag{121}$$

  So, in this case,

  - the **phase constant** β is the same as the **wavenumber** k **for a lossless medium**,
  - the **intrinsic impedance** η_c is the same as the **intrinsic impedance** η **of a lossless medium**.

- **Good Conductors**

  In this case, $\epsilon''/\epsilon' > 100$ and

$$\alpha \approx \omega\sqrt{\frac{\mu\epsilon''}{2}} = \omega\sqrt{\frac{\mu\sigma}{2\omega}} = \sqrt{\pi f \mu\sigma} \quad \text{(Np/m)}, \tag{122}$$

$$\beta \approx \alpha \approx \sqrt{\pi f \mu\sigma} \quad \text{(rad/m)}, \tag{123}$$

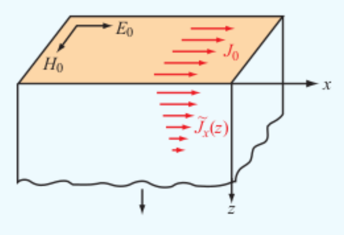
$$\eta_c \approx \sqrt{j\frac{\mu}{\epsilon''}} = (1+j)\sqrt{\frac{\pi f \mu}{\sigma}} \approx (1+j)\frac{\alpha}{\sigma} \quad (\Omega). \tag{124}$$

- We explored an example of finding $\mathbf{E}(z,t)$ and $\mathbf{H}(z,t)$ and the depth z where $|\mathbf{E}(z)| = 0.01|\mathbf{E}(0)|$ for a uniform plane wave in seawater.

Current Flow in a Good Conductor

- The **current** associated with a **DC voltage source** is **evenly distributed** throughout a conductor.
- In contrast, the **current** associated with an **AC voltage source** mostly **flows through a layer near the surface**.

Consider a semi-infinite conductor bounded by a perfect dielectric at the x-y plane with the z axis pointing into the conductor.



(a) Exponentially decaying $\tilde{J}_x(z)$

- An x-**polarized electric field**

$$\tilde{\mathbf{E}} = E_0 \hat{\mathbf{x}} \qquad (125)$$

at the surface

induces a plane wave

$$\tilde{\mathbf{E}}(z) = E_0 e^{-\alpha z} e^{-j\beta z} \hat{\mathbf{x}} \tag{126}$$

$$\tilde{\mathbf{H}}(z) = \frac{E_0}{\eta_c} e^{-\alpha z} e^{-j\beta z} \hat{\mathbf{y}} \tag{127}$$

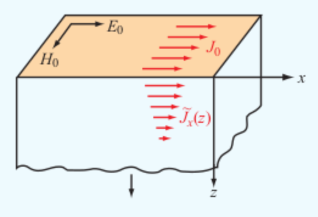
in the conductor. This creates a current

$$\tilde{\mathbf{J}}(z) = J_0 e^{-(1+j)z/\delta_s} \hat{\mathbf{x}} \ (\mathrm{A/m^2}), \tag{128}$$

where

$$J_0 = \sigma E_0 \tag{129}$$

is the **current density in the conductor at the boundary**.



(a) Exponentially decaying $\tilde{J}_x(z)$

The **skin depth**

$$\delta_s = \frac{1}{\alpha} = \frac{1}{\sqrt{\pi f \mu \sigma}} \text{ (m).} \qquad (130)$$

- The **total current passing through a slice of the conductor with width w in the y direction** is

$$\tilde{I} = \frac{J_0 w \delta_s}{1+j}. \qquad (131)$$

This is an excellent approximation if the depth is finite > a few times δ_s.

- If the slice has thickness ℓ in the x direction, the **voltage across the slice at the surface** is

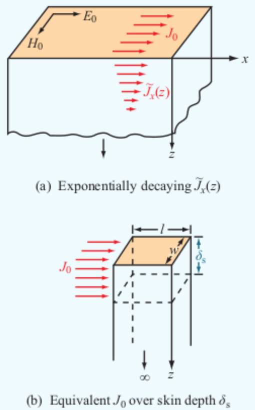


Figure 7-15 Exponential decay of current density $\tilde{J}_x(z)$ with z in a solid conductor. The total current flowing through (a) a section of width w extending between $z = 0$ and $z = \infty$ is equivalent to (b) a constant current density J_0 flowing through a section of depth δ_s.

$$\tilde{V} = E_0 \ell = \frac{J_0}{\sigma} \ell. \qquad (132)$$

- The **impedance of the slice** is

$$Z = \frac{\tilde{V}}{\tilde{I}} = Z_s \frac{\ell}{w} \; (\Omega), \qquad (133)$$

where

$$Z_s = \frac{1 + j}{\sigma \delta_s} \qquad (134)$$

is the **internal** or **surface impedance**.

- Here,

$$Z_s = R_s + j\omega L_s \qquad (135)$$

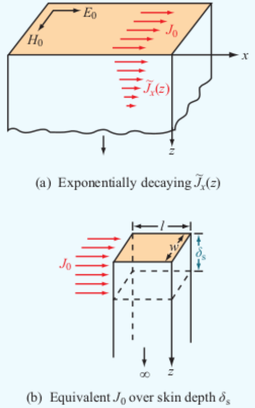


Figure 7-15 Exponential decay of current density $\tilde{J}_x(z)$ with z in a solid conductor. The total current flowing through (a) a section of width w extending between $z = 0$ and $z = \infty$ is equivalent to (b) a constant current density J_0 flowing through a section of depth δ_s.

with **surface resistance**

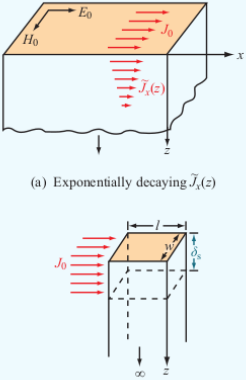
$$R_s = \sqrt{\frac{\pi f \mu}{\sigma}} \; (\Omega) \qquad (136)$$

and inductance

$$L_s = \frac{1}{2}\sqrt{\frac{\mu}{\pi f \sigma}} \; (\mathrm{H}). \qquad (137)$$

- The **ac resistance of a slab of width** w **and length** ℓ is

$$R = R_s \frac{\ell}{w}. \qquad (138)$$



(a) Exponentially decaying $\tilde{J}_x(z)$

(b) Equivalent J_0 over skin depth δ_s

Figure 7-15 Exponential decay of current density $\tilde{J}_x(z)$ with z in a solid conductor. The total current flowing through (a) a section of width w extending between $z = 0$ and $z = \infty$ is equivalent to (b) a constant current density J_0 flowing through a section of depth δ_s.

- The **ac resistance of a coaxial cable per unit length** is

$$R' = \frac{R_s}{2\pi}\left(\frac{1}{a} + \frac{1}{b}\right) \ (\Omega/\mathrm{m}), \quad (139)$$

where R_s is the surface resistance of the conducting material, a is the radius of the inner conductor, b is the radius of the outer conductor, and both conductors are at least several times the skin depth. (The surface resistance and skin depth depend on the frequency of the wave.)

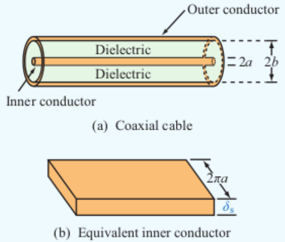


Figure 7-16 The inner conductor of the coaxial cable in (a) is represented in (b) by a planar conductor of width $2\pi a$ and depth δ_s, as if its skin has been cut along its length on the bottom side and then unfurled into a planar geometry.

Electromagnetic Power Density

- The **Poynting vector** is

$$\mathbf{S} = \mathbf{E} \times \mathbf{H} \quad (\mathrm{W/m^2}), \tag{140}$$

specifies the instantaneous power of an EM wave in the direction $\hat{\mathbf{k}}$ the wave propagates.

- The **power flowing through an area** A **with unit surface vector** $\hat{\mathbf{n}}$ is

$$P = \int_A \mathbf{S} \cdot \hat{\mathbf{n}}\, dA = SA\cos\theta \quad (\mathrm{W}), \tag{141}$$

where $S = |\mathbf{S}|$ and θ is the angle between $\hat{\mathbf{k}}$ and $\hat{\mathbf{n}}$.

- The **average power density**

$$\mathbf{S}_{\mathrm{av}} = \frac{1}{2}\mathrm{Re}\left\{\tilde{\mathbf{E}} \times \tilde{\mathbf{H}}^*\right\} \quad (\mathrm{W/m^2}) \tag{142}$$

is a better measure of the power the wave carries.

Review for Exam 1 (Continued)

- In a **lossless medium**,

$$\mathbf{S}_{\text{av}} = \frac{1}{2\eta}|\tilde{\mathbf{E}}|^2\hat{\mathbf{k}} \quad (\text{W/m}^2). \tag{143}$$

- In a **lossy medium**,

$$\mathbf{S}_{\text{av}} = \frac{|\tilde{\mathbf{E}}(0)|^2}{2|\eta_c|}e^{-2\alpha z}\cos\theta_\eta\hat{\mathbf{z}} \quad (\text{W/m}^2), \tag{144}$$

  where

$$\eta_c = |\eta_c|e^{j\theta_\eta}. \tag{145}$$

- In a lossy medium, the **power decreases at twice the rate** that the electric and magnetic fields attenuate.
- A power ratio P_1/P_2 is often expressed in **decibels**:

$$G\ (\text{dB}) = 10\log\left(\frac{P_1}{P_2}\right) = 20\log\left(\frac{V_1}{V_2}\right). \tag{146}$$

Here, V_1 and V_2 are voltages applied across the same resistor and P_1 and P_2 are the power dissipated in each case, respectively.

- Decibels are **usually used** to compare a **measured power** P_{meas} to a **reference power** P_0.
- The rate of decrease of $S_{\text{av}}(z)$ is the **attenuation rate**

$$A = 10 \log \left(\frac{S_{\text{av}}(z)}{S_{\text{av}}(0)} \right) = -8.69 \left(\alpha \ (\text{Np/m}) \right) z = -\left(\alpha \ (\text{dB/m}) \right) z, \tag{147}$$

where

$$\alpha \ (\text{dB/m}) \equiv 8.69\alpha \ (\text{Np/m}). \tag{148}$$

- The **attenuation rate** can be written in terms of the **magnitude of the electric field**

$$A = 20 \log \left(\frac{|E(z)|}{|E(0)|} \right) \quad (\text{dB}). \tag{149}$$

Homework

Homework 2 is due on Tuesday at the beginning of class.

Exam 1 will be a week from today, on Thursday, May 7$^{\text{th}}$.